Empirical priors for inference on high-dimensional structured parameters[1]

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Introduction

- High-dimensional problems are everywhere now.
- Hopeless without some structural assumptions.
- *Sparsity* is a common type of structure, i.e., most signals are zero/small but a few are large.
- An assumed structure is "prior information," so a Bayesian approach is natural/convenient.
- However:
  - no real information about structure-specific parameters;
  - it's high-dim, so the prior on these parameters matters!

Intro, cont.

- The tails on the structure-specific prior play a role:
  - thin-tailed conjugate priors are good for computation but lead to sub-optimal convergence properties
  - heavy-tailed priors are good theoretically but harder to compute with.
- Do we have to sacrifice one for the other?
- *Key insight:* Prior tails don't matter if it's correctly centered.
- "Correctly centered" requires the data — an *empirical prior*.
- This is intuitively reasonable, but how to make it work in theory and in practice?

This talk

- Empirical priors for a sparse normal mean
- Other problems involving sparsity
  - high-dim regression
  - high-dim precision matrix
- Empirical priors for a piecewise polynomial structure
  - problem setup
  - posterior concentration properties
  - numerical results
- Concluding remarks

Sparse normal means

- $Y_i \overset{\text{ind}}{\sim} \mathsf{N}(\theta_i, \sigma^2)$, $i = 1, \ldots, n$.
- High-dim, since each Y_i has its own θ_i.
- Assume the vector θ is *sparse*, i.e., most θ_i's are 0.
- Express θ vector as a pair (S, θ_S):
  - $S \subseteq \{1, 2, \ldots, n\}$ denotes the location of non-zeros
  - θ_S the $|S|$-vector of non-zero values.
- Need a prior for (S, θ_S)...
- Sparsity suggests a marginal prior $\pi_n(S)$ for S:
  - $|S| \sim f(s) \propto n^{-as}$, $s = 0, \ldots, n$, where $a > 0$.
  - Given the size, S is uniform over all configs of that size.

Sparse normal means, cont.

- Conditional prior for θ_S, given S?
- Instead of a heavy-tailed (e.g., Laplace) prior centered at 0, let's take a conjugate normal prior centered at the data:

$$\pi_n(\theta_S \mid S) = \mathsf{N}_{|S|}(Y_S, \gamma^{-1} I_{|S|}), \quad \gamma \in (0, 1).$$

- Conditional prior for θ_S, given S, times the marginal prior for S gives an *empirical prior* for θ — call it Π_n.
- Intuition:
  - informative prior on the thing we know about, S;
  - "non-informative" on the thing we don't know about, θ_S

Sparse normal means, cont.

- Combine prior and likelihood in *almost* the usual way:

$$\Pi^n(d\theta) \propto L_n(\theta)^\alpha \, \Pi_n(d\theta), \quad \alpha \in (0,1).$$

- I'd be happy to answer questions about the power α.
- Relatively simple computations thanks to conjugacy.
- Good theoretical convergence properties too:[2]
  - optimal concentration rate $|S^\star| \log(n/|S^\star|)$
  - learns structure $S^\star$ if $\min_{i \in S^\star} |\theta_i^\star| \gtrsim (\log n)^{1/2}$
  - valid uncertainty quantification (on linear functionals)

[2] M. and Walker (2014, `arXiv:1304.7366`); M. and Ning (2020, `arXiv:1812.02150`)

More sparsity cases

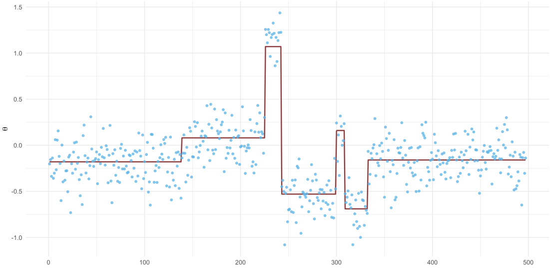
- Linear regression
  - Not too much different from the means model
  - Lots of good theory and numerical results[3]
  - New variational approximation with nice properties[4]
- Gaussian graphical models
  - Sparsity corresponds to conditional independence between pairs of variables, characterized by an underlying graph.
  - We proposed a new *empirical G-Wishart* prior,[5] and show very good theoretical properties and empirical performance.
  - Dr. Chang Liu is presenting this work in JSM Session 578.

[3] M., et al (2017, `arXiv:1406.7718`; 2020, `arXiv:1903.00961`)
[4] Yang and M. (2020, `arXiv:2007.15930`)
[5] Liu and M. (2019, `arXiv:1912.03807`)

- Consider the same normal means model

$$Y_i \overset{\text{ind}}{\sim} \mathsf{N}(\theta_i, \sigma^2), \quad i = 1, \ldots, n.$$

- This time, however, the vector θ isn't sparse; instead, it has a piecewise polynomial structure.[6]
- Plot shows piecewise constant — my focus here.



[6] *NEW* Liu, M., and Shen (2020, `arXiv:1712.03848`)

Piecewise, cont.

- The mean vector can be described via:
  - a block configuration B (a simple partition)
  - a vector θ_B of block-specific parameters
- This is not-too-complicated in the piecewise constant case; basically, for fixed B, it's analysis of variance.
- More complicated for general piecewise polynomial...
- Same idea from before can be applied here:
  - use a low-complexity assumption to construct a prior for B
  - let the data inform prior for B-specific parameters θ_B.

Piecewise, cont.

- Piecewise constant case only here!
- Marginal prior $\pi_n(B)$ for B:
  - $|B| \sim f_n(b) \propto n^{-\lambda(b-1)}$, $b = 1, \ldots, n$
  - uniform on all B of given size $|B|$.
- Conditional prior for θ_B, given B:

$$\theta_{B(s)} \overset{\text{ind}}{\sim} \mathsf{N}(\hat{\theta}_{B(s)}, v|B(s)|^{-1}), \quad s = 1, \ldots, |B|,$$

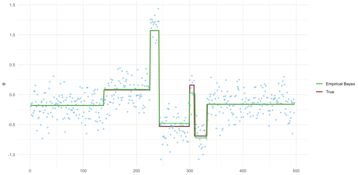
  where the $\hat{\theta}_{B(s)}$'s are from ANOVA.
- Combine the induced prior Π^n for $\theta = (B, \theta_B)$ with (α power of likelihood) to get a posterior Π^n.

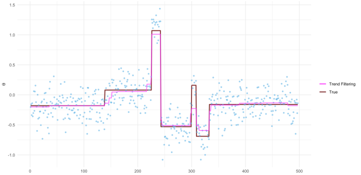
Piecewise, cont.

- Summary of the theoretical properties.
- Concentration rate:
  - $\mathsf{E}_{\theta^\star} \Pi^n(\{\theta \in \mathbb{R}^n : \|\theta - \theta^\star\|^2 \gtrsim |B_{\theta^\star}| \log n\}) \to 0.$
  - Matches the minimax optimal rate except for in high-complexity cases where, e.g., $|B_{\theta^\star}| = O(n/\log n)$.
- Structure learning:
  - Identifiability of $B_{\theta^\star}$ is subtle in general piecewise polynomial cases, but it's easy for piecewise constant.
  - $\mathsf{E}_{\theta^\star} \pi^n(B_{\theta^\star}) \to 1$ if, among other things,

$$(\text{min block size of } \theta^\star) \times (\text{min jump size in } \theta^\star)^2 \gtrsim \log n.$$

  - Conditions match the best I've seen in the change-point detection literature (Fryzlewicz, *Annals* 2014).

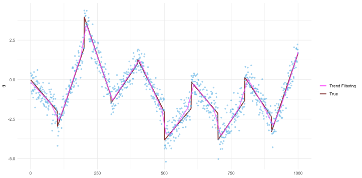




- Compare proposed method (left) with trend filtering (right).
- Over 1000 simulated data sets; $|B^\star| = 7$.

Method	$\mathsf{E}\|\hat{\theta} - \theta^\star\|^2$	$\mathsf{E}\|\hat{B}\|$	$\mathsf{P}(\hat{B} = B^\star)$	$\mathsf{P}(\hat{B} \supset B^\star)$	$\mathsf{P}(\hat{B} \subset B^\star)$
EB	**0.70**	**7.01**	**0.19**	0.19	**0.19**
TF	1.52	8.77	0.05	**0.28**	0.07

Piecewise, cont.



- Compare proposed method (left) with trend filtering (right).
- Estimation error: 15 for EB and 30 for TF.
- Based on simulations in the paper:
  - EB outperforms TF when signal is “discontinuous” or “continuous but not smooth.”
  - TF outperforms EB when signal is “smooth.”

Concluding remarks

- Empirical priors are promising: retain conjugacy without sacrificing on concentration properties.
- They can be viewed as "non-informative"...
- New results here for piecewise polynomial structures.
- Focus here was on Gaussian problems, but those aren't the only possible applications:
  - general theory paper[7] considers a number of examples
  - one application to monotone density estimation[8]
- Two problems I'm currently interested in are:
  - general mixture models
  - sparse GLMs

[7] M. and Walker (2019, `arXiv:1604.05734`)
[8] M. (2019, `arXiv:1706.08567`)

The end... except for a commercial

Thank you!

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